

Xin YANG

AI-based Multisensing & Data Fusion | Condition-based Maintenance |
Structural Health Monitoring

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ABOUT ME

My research focuses on damage identification and reliability assessment of composite materials and lightweight structures, integrating [ultrasonic guided waves](#), [multisensor fusion](#), and [artificial intelligence](#). My primary research interests include damage imaging under sparse observations, probabilistic [fusion of heterogeneous sensing information](#), and [trustworthy diagnostics under limited-data and cross-domain conditions](#). I also collaborate on [physics-informed learning](#) for structural parameter identification and mechanical response modelling.

I have published research in journals including **Mechanical Systems and Signal Processing**, **International Journal of Mechanical Sciences**, **Composite Structures**, and **Reliability Engineering & System Safety**. I am currently a postdoctoral researcher at the Faculty of Aerospace Engineering, Delft University of Technology, where I work on multimodal sensing and AI-driven health monitoring of polymer bearing cages.

PROFESSIONAL EXPERIENCE

2026.06 - Present	Postdoctoral Researcher	TU Delft, Netherlands
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- Developing multimodal sensing approaches based on [Fibre Bragg Grating sensors](#), [acoustic emission](#), and [thermal imaging](#) to monitor deformation, fatigue damage, and failure in bearing cages, while drawing on ultrasonic testing expertise to identify damage-sensitive features.
- Investigating [fusion methods](#) for heterogeneous time-series signals and imaging data, with an emphasis on multichannel data acquisition, condition identification, and nondestructive monitoring.

2023.02 - 2026.08	PhD Researcher	KU Leuven, Belgium
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- Investigated [optimal sensor placement](#) and [multisensor fusion](#) at the data, feature, and decision levels for the structural health monitoring of aerospace composite structures.

- Developed [conditional diffusion models](#) for damage imaging, probabilistic fusion of guided-wave and vibration data, Transformer-based multimodal modelling, and uncertainty quantification for structural diagnostics.

EDUCATION

2023.02-2026.08	KU Leuven, Belgium	Mechanical Engineering	Ph.D.
2025.12-2026.01	University of Bristol, UK	Mechanical Engineering	Visiting Ph.D.
2019.09-2022.03	Tongji University, CN	Vehicle Engineering	MEng.
2015.09-2019.06	Chongqing University, CN	Vehicle Engineering	BEng.

RESEARCH PROJECTS

- **20+ peer-reviewed publications** in top-tier journals and international conferences, including *Mechanical Systems and Signal Processing* (IF: 10.2), *Reliability Engineering & System Safety* (IF: 13.7), and related venues in industrial health monitoring and intelligent sensing.
- **Active peer reviewer** for leading journals in the field, contributing to the editorial evaluation of research in signal processing, machine learning for condition monitoring, and sensor-based diagnostics.